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Convergent adaptive finite elements for the nonlinear Laplacian

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Summary. The numerical solution of the homogeneous Dirichlet problem for the p-Laplacian, $p \in]1, \infty[$, is considered. We propose an adaptive algorithm with continuous piecewise affine finite elements and prove that the approximate solutions converge to the exact one. While the algorithm is a rather straight-forward generalization of those for the linear case $p = 2$, the proof of its convergence is different. In particular, it does not rely on a strict error reduction.

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1 Introduction and main result

The *adaptive* numerical solution of elliptic boundary value problems has become very popular. Adaptivity is indeed an effective tool for obtaining a good quality of approximate solutions at relatively low computational costs. In particular, it is especially successful in the presence of singularities. For a survey, we refer to the books of Ainsworth and Oden [1] and of Verfürth [23].

Adaptive methods are *iterative*: the current approximate solution is used to correct (or adapt) the discretization for a new approximate solution. Thus, a typically infinite sequence of approximate solutions is generated and the question arises *whether this sequence converges to the exact solution*. In the case of linear problems, this question has been answered affirmatively; see

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Babuška and Vogelius [2] (in 1d), Morin *et al.* [19,20] (multidimensional), as well as Dörfler [13] and Dörfler and Wilderotter [15] for closely related work. The nonlinear case seems to be quite open; a work in this direction is Dörfler [12].

This article concerns the convergence of an adaptive algorithm for the following nonlinear problem: let $p \in]1, \infty[$ and find $u : \Omega \subset \mathbb{R}^d \to \mathbb{R}$ such that

$$-\mathrm{div}\left(|\nabla u|^{p-2}\nabla u\right) = f \quad \text{in } \Omega, \qquad u = 0 \quad \text{on } \partial\Omega. \tag{1.1}$$

Due to the degenerate character of the operator or the geometry of $\partial\Omega$, the solution may not be smooth even for smooth f. Consequently, it is advantageous to apply adaptive methods. The algorithm, which is basically a straight-forward generalization of those in [19,20], uses continuous piecewise affine finite elements. It consists of the following main steps in every iteration:

$$\text{Solve} \;\to\; \text{Estimate} \;\to\; \text{Mark} \;\to\; \text{Refine}. \tag{1.2}$$

We comment on some aspects of these steps. Doing so, we restrict to the case $d = 2$ regarding the terminology for the triangulation.

In step 'Solve', the finite element solution u_h is computed. As in [19, 20], we suppose that the involved integrals can be calculated exactly. This allows us to concentrate on the difficulties caused by the nonlinear operator in Problem (1.1).

In step 'Estimate', *local indicators* are computed. They provide the information for adapting the discretization. Two indicator types are used: *hierarchical error indicators* and *data oscillation indicators*. The hierarchical error indicators are related to the residual with respect to (1.1) of the approximate solution u_h. More precisely, they test the residual against certain discrete functions over a finer triangulation and they fit (up to some scaling) into the framework of [23, Propositions 1.14 and 1.15] in the linear case $p = 2$. Furthermore, they exhibit the following crucial property: together with the data oscillation indicators, the hierarchical error indicators yield a global upper bound of the residual of u_h in the dual energy norm. Data oscillation measures information missed in testing the residual with (a finite number of) discrete functions. Its importance in adaptivity is discussed in Morin *et al.* [19,20] and Dörfler and Nochetto [14], while its connection with a posteriori error estimates is illustrated in Verfürth [23, Sections 1.1 and 2.3].

In step 'Mark', triangles and sides that will be divided are selected (or *marked*) by comparing indicator values. Thereby, we employ the *fixed fraction criterion* in ℓ_{p^*} — a marking principle that was proposed by Dörfler [13] in the case $p = 2$.

In step 'Refine', new degrees of freedom are created by refining in such a way that

- the minimum angle of the new triangulation is bounded away from 0 in terms of the minimum angle of the initial triangulation,
- the new finite element space contains the old one,
- the new finite element space contains the discrete test functions belonging to marked indicators.

A condition like the last one was first introduced by Dörfler [13].

If one marks all triangles and all sides in every iteration, then the indicated algorithm leads to consecutive uniform refinements and the maximal diameter $h_{\max}$ of a triangle decreases to 0. This non-adaptive case is well investigated (see Barrett and Liu [4], Chow [8], Glowinski and Marrocco [16], Wei [24], and for more general right-hand sides, Jebelean [18]) and it is well known that the sequence of approximate solutions converges to the exact solution. The latter is proved with the help of $h_{\max} \searrow 0$; cf. e.g. [16, Théorème 4.1]. However, in general, $h_{\max} \searrow 0$ is not desirable in the adaptive case and even not necessarily true. In fact, there are (linear) 1d examples for which $h_{\max}$ stays bounded away from 0; see Babuška and Vogelius [2, Remark 6.1]. Nevertheless:

Main result. Let u be a weak solution of Problem (1.1) and denote by $(u_l)_l$ the sequence of approximate solutions generated through iterating (1.2). Then

$$u_l \to u \quad \text{in} \quad W_p^1(\Omega) \quad \text{as} \quad l \to \infty.$$

The hierarchical error indicators may be replaced by equivalent (up to data oscillation) error indicators. The algorithm still converges, if the new finite element space contains the discrete test functions belonging to those hierarchical indicators that are needed to bound the marked error indicators. As a consequence, one may use the residual error indicators appearing in Baranger and El Amri [3, Sect. 2] (and Verfürth [23, Proposition 3.12]).

We compare the afore-mentioned result and its proof with the one in [19] for the case $p = 2$. In [19], a linear convergence rate for the energy norm is established. An essential ingredient for that is a meaningful improvement of the *error reduction theorem* in [13]. The improvement states that one adaptive iteration (1.2) reduces the error by a *fixed* fraction provided data oscillation is relatively small (in the following discussion we neglect this condition for simplicity). It is proved with the help of two auxiliary results: first, the orthogonality result

$$\|u_{l+1} - u\|_\Omega^2 \;=\; \|u_l - u\|_\Omega^2 \;-\; \|u_l - u_{l+1}\|_\Omega^2 \,, \tag{1.3}$$

where $\|\cdot\|_\Omega$ denotes the energy norm; second, a lower bound of the error reduction $\|u_l - u_{l+1}\|_\Omega$ in terms of the old error $\|u_l - u\|_\Omega$. The latter estimation consists of two parts. The first part is a global variant of the standard

a posteriori lower bound: the error reduction $|||u_l - u_{l+1}|||_\Omega$ is bounded from below by the ℓ_2-sum of all local error indicators. This is due to the third point in the description of step 'Refine' and the fixed fraction criterion. The second part is the standard a posteriori upper bound for the error $|||u_l - u|||_\Omega$ in terms of all local error indicators. In order to combine the two parts, it is crucial that the quotient of the two bounds depends only on the shape-regularity of the underlying triangulation.

Unfortunately, in the nonlinear case $p \neq 2$, the afore-mentioned quotient does not have this nice property; we refer to Verfürth [23, Proposition 3.12]. The reason behind this is a gap between the ellipticity and continuity properties of the p-Laplacian in the energy norm; see e.g. Glowinski and Marrocco [16, Sect. 5] or Remark 3.5. Moreover, the orthogonality relationship (1.3) hinges on linearity.

Therefore, we invoke the fact that Problem (1.1) can be rewritten as a minimization of a suitable energy functional J. Since the discrete spaces are nested, we derive that the *energy* reduction $J(u_l) - J(u_{l+1})$, which coincides with the squared *error* reduction $\frac{1}{2}|||u_{l+1} - u_l|||^2_\Omega$ in the linear case $p = 2$, decreases to 0 as $l \to \infty$. Consequently, we may show convergence by estimating the dual energy norm of the residual $DJ(u_l)$, which coincides with $|||u_l - u|||_\Omega$ for $p = 2$, in terms of the energy reduction $J(u_l) - J(u_{l+1})$. For this estimate, we use the third point in the description of step 'Refine' and the fixed fraction criterion. The afore-mentioned gap is no problem in the indicated procedure. However, this procedure does not lead to a linear convergence rate.

The rest of this article is organized as follows: in Sect. 2 we recall the weak and variational formulation of Problem (1.1) and introduce its discretization as well as the adaptive algorithm. The main purpose of this article, namely that this algorithm provides a convergent sequence of approximate solutions, is tackled in Sect. 3. We conclude that section by discussing the convergence properties of alternative local error indicators.

2 Continuous problem and its adaptive approximation

We first recall the weak and variational formulation of the homogeneous Dirichlet problem for the p-Laplacian. Relying on these formulations, we then discretize with continuous linear finite elements. Furthermore, various computable quantities are introduced. They serve to guide the adaptive algorithm. We finally formulate this algorithm.

2.1 Continuous problem

Let Ω be a bounded, open, connected, polyhedral set in $\mathbb{R}^d$, $d \in \{1, 2, 3\}$, with Lipschitz boundary $\partial\Omega$. Let $p \in]1, \infty[$ and denote by $L_p(\Omega)$ the space of Lebesgue-measurable functions that are p-integrable. We write $\|\cdot\|_{p;\Omega}$ for its norm. The dual exponent of p is $q := p^* := p/(p-1)$.

It is convenient to introduce the continuous vector field

$$a : \mathbb{R}^d \to \mathbb{R}^d \quad \text{with} \quad a(\xi) := \begin{cases} |\xi|^{p-2}\xi, & \text{if } \xi \neq 0, \\ 0, & \text{if } \xi = 0. \end{cases}$$

This vector field satisfies the following two inequalities (see e.g. Barrett and Liu [5, Lemma 2.2]): for $\xi_1, \xi_2 \in \mathbb{R}^d$ (with $(\xi_1, \xi_2) \neq 0$ if $p \in]1, 2[$) holds

$$|a(\xi_1) - a(\xi_2)| \;\leq\; C_1 \left(|\xi_1| + |\xi_2|\right)^{p-2} |\xi_1 - \xi_2|, \tag{2.1a}$$

$$[a(\xi_1) - a(\xi_2)] \cdot (\xi_1 - \xi_2) \;\geq\; C_2 \left(|\xi_1| + |\xi_2|\right)^{p-2} |\xi_1 - \xi_2|^2, \tag{2.1b}$$

where $C_i > 0$, $i = 1, 2$, depend only on p and d. Applying the triangle inequality to the second factor in the right-hand side of (2.1a) yields

$$|a(\xi_1) - a(\xi_2)| \;\leq\; C_1 \, |\xi_1 - \xi_2|^{p-1}, \tag{2.2}$$

i.e. the vector field a is globally Hölder continuous, if $p \in]1, 2]$. Similarly, if $p \geq 2$, one obtains

$$[a(\xi_1) - a(\xi_2)] \cdot (\xi_1 - \xi_2) \;\geq\; C_2 \, |\xi_1 - \xi_2|^p, \tag{2.3}$$

which states that a is uniformly monotone in this case.

Let $f \in L_q(\Omega)$ and define

$$V := \{w \in W_p^1(\Omega) \mid w_{|\partial\Omega} = 0\},$$

where $W_p^1(\Omega)$ denotes the Sobolev space of $L_p(\Omega)$-functions that have weak derivatives in $L_p(\Omega)$.

The following weak formulation of (1.1) is obtained, as usual, by means of integration by parts.

Problem 2.1 (Weak formulation) *Find $u \in V$ such that*

$$\forall\, \varphi \in V \qquad \int_\Omega a(\nabla u) \cdot \nabla\varphi = \int_\Omega f\varphi.$$

We state some well-known facts about Problem (2.1). For their proofs, one may consult Glowinski and Marrocco [16] or the account of Ciarlet [9, Section 5.3]. Problem (2.1) is well-posed and the solution u satisfies

$$\|\!|u|\!\|_\Omega := \|\nabla u\|_{p;\Omega} = \|\!|f|\!\|_{\Omega;*}^{1/(p-1)}, \tag{2.4}$$

where

$$\|\!|f|\!\|_{\Omega;*} := \sup\{\langle f, \varphi\rangle \mid \varphi \in V \text{ with } \|\!|\varphi|\!\|_\Omega \le 1\}.$$

Note that, thanks to a Poincaré inequality, $\|\!|\cdot|\!\|_\Omega$ and the norm of $W_p^1(\Omega)$ are equivalent in V. Problem (2.1) is equivalent to the following minimization problem, which involves the strictly convex functional $J : V \to \mathbb{R}$ with

$$J(v) := \int_\Omega \frac{1}{p}|\nabla v|^p - \int_\Omega fv.$$

Problem 2.2 (Variational formulation) *Find $u \in V$ such that*

$$\forall\, v \in V \quad J(u) \le J(v).$$

This formulation will be very useful for our purposes.

Notice that we do not suppose that the domain Ω is smooth or convex. Accordingly, the geometry of the boundary $\partial\Omega$ may cause singularities of the solution u already in the linear case $p = 2$; see e.g. Grisvard [17]. In addition, the degenerate character of the p-Laplacian for $p \neq 2$ may lead to $u \notin W^{2,p}(\Omega)$; see [16, Section 8.1].

2.2 Discrete problem

Let $\mathcal{T}_h$ be a triangulation (in the sense of [7, (3.3.11)]) of Ω. Given a (closed) simplex $T \in \mathcal{T}_h$, we write h_T for its diameter and ρ_T for the maximal radius of a ball that is contained in T. The *shape-regularity* of the triangulation $\mathcal{T}_h$ is defined by

$$\gamma(\mathcal{T}_h) := \max_{T\in\mathcal{T}_h} \frac{h_T}{\rho_T} \in]1, \infty[.$$

In the case of $d = 2$, it is related to the minimum angle of $\mathcal{T}_h$; see e.g. [9, Exercise 3.1.3]. We write $\mathcal{N}_h$ for the set of interior nodes and $\mathcal{S}_h$ for the set of the interelement sides of $\mathcal{T}_h$.

Let V_h denote the finite-dimensional space of continuous piecewise affine finite elements over $\mathcal{T}_h$ with zero boundary values, that is

$$V_h := V(\mathcal{T}_h) := \{v_h \in C^0(\overline{\Omega}) \mid v_{h|\partial\Omega} = 0 \text{ and } \forall\, T \in \mathcal{T}_h \;\; v_{h|T} \in P_1(T)\} \subset V.$$

Here, $P_1(T)$ denotes the restriction to T of the space of all polynomials in $\mathbb{R}^d$ with degree equal or less than 1. Let $(\lambda_h^z)_{z\in\mathcal{N}_h}$ be the hat functions that are defined by $\lambda_h^z \in V_h$, $\lambda_h^z(z) = 1$ and $\lambda_h^z(\bar{z}) = 0$ for all $\bar{z} \in \mathcal{N}_h \setminus \{z\}$.

In order to approximate solutions of Problem (2.1) we use

Problem 2.3 (Discrete weak formulation) *Find $u_h \in V_h$ such that*

$$\forall\, \varphi_h \in V_h \qquad \int_\Omega a(\nabla u_h)\cdot\nabla\varphi_h = \int_\Omega f\varphi_h.$$

As before, we state some well-known facts and refer to [16] or the account in [9, Section 5.3] for proofs. Problem (2.3) is well-posed and the solution u_h satisfies the *stability estimate*

$$\|\!|u_h|\!\|_\Omega \le \|\!|f|\!\|_{\Omega;*}^{1/(p-1)}. \tag{2.5}$$

Moreover, Problem (2.3) is equivalent to the following minimization problem, which is the discrete counterpart of Problem (2.2).

Problem 2.4 (Discrete variational formulation) *Find $u_h \in V_h$ such that*

$$\forall\, v_h \in V_h \quad J(u_h) \le J(v_h).$$

For the numerical solution of Problem (2.3) (or Problem (2.4)), we refer to [4, Section 5] and [16, Section 9].

2.3 Indicators for the error and data oscillation

The local error indicators are indexed by members of a certain set $\mathcal{K}_h$. If $d = 1$, we take the set of all triangles: $\mathcal{K}_h := \mathcal{T}_h$. Otherwise, we take the set of all interelement sides and triangles in $\mathcal{T}_h$: $\mathcal{K}_h := \mathcal{S}_h \cup \mathcal{T}_h$. We define the local error indicators with the help of hat functions belonging to a finer triangulation than $\mathcal{T}_h$. To obtain this finer triangulation, we perform a fixed number of bisections to each simplex of $\mathcal{T}_h$ such that

$$\forall\, K \in \mathcal{K}_h \;\; \exists\, z \in \mathcal{N}_{h/2} \setminus \mathcal{N}_h \quad z \in K,$$

holds. Hereafter, $\mathcal{N}_{h/2}$ is the set of all interior nodes of the resulting finer triangulation $\mathcal{T}_{h/2}$. For $d = 1, 2, 3$, we perform 1,3,6 bisections, respectively, to each simplex. By construction, we have

$$V_h \;\subset\; V_{h/2} \;:=\; V(\mathcal{T}_{h/2}).$$

In other words: $V_{h/2}$ is in between V_h and V and for every element of $\mathcal{K}_h$ there is an additional node. Let $\lambda_{h/2}^z$ be the hat function in $V_{h/2}$ that

corresponds to the node $z \in \mathcal{N}_{h/2}$. For each $K \in \mathcal{K}_h$ we choose some fixed $z_K \in \big(\mathcal{N}_{h/2} \setminus \mathcal{N}_h\big) \cap K$ and set

$$\varphi_h^K := \frac{\lambda_{h/2}^{z_K}}{\|\lambda_{h/2}^{z_K}\|_\Omega}.$$

We then associate with each $K \in \mathcal{K}_h$ the *local error indicator*

$$\eta_h(K) := \left| \int_\Omega a(\nabla u_h) \cdot \nabla \varphi_h^K - \int_\Omega f \varphi_h^K \right|, \tag{2.6}$$

which tests the residual of the discrete solution u_h with φ_h^K. Its global counterpart η_h is obtained as "ℓ_q-sum":

$$\eta_h := \left(\sum_{K \in \mathcal{K}_h} \eta_h(K)^q \right)^{1/q}.$$

Apart from $\eta_h(K)$, $K \in \mathcal{K}_h$, and η_h, we need the so-called *data oscillation*. It is denoted by δ_h. Similar to η_h, it splits into local contributions in a "ℓ_q-sense":

$$\delta_h := \left(\sum_{T \in \mathcal{T}_h} \delta_h(T)^q \right)^{1/q} \quad \text{with} \quad \delta_h(T) := h_T \, \|f_T - f\|_{q;T}, \tag{2.7}$$

where f_T denotes the mean value of f in T: $f_T := |T|^{-1} \int_T f$.

Remark 2.1 Discrete functions similar to $\lambda_{h/2}^{z_K}$, $K \in \mathcal{K}_h$, are used in [13, 19, 20] as to obtain local lower bounds for the error reduction of one adaptive iteration.

*Remark 2.2 (*Normalization) Discrete test functions like φ_h^K, $K \in \mathcal{K}_h$, that are normalized in the energy norm are used in Bornemann *et al.* [6, Section 3], Deuflhard *et al.* [11], and Petzold [21]. Such discrete test functions lead to "constant free" local lower bounds with respect to the energy error for $p = 2$. Here, we have a "constant free" local lower bound with respect to the residual in the dual energy norm:

$$\eta_h(K) \leq \|DJ(u_h)\|_{\operatorname{supp} \varphi_h^K;*}. \tag{2.8}$$

Hereafter, $DJ(u_h)$ denotes the Gateaux derivative of J in u_h.

Remark 2.3 The notion of data oscillation in (2.7) is a natural generalization of the one in [19] in the context at hand. See also [23, Section 3.3].

2.4 Adaptive algorithm

We formulate the main steps of the adaptive algorithm. Thereafter, we replace the subscript "h" by an iteration counter called "l".

Algorithm 2.1. Let $\theta \in]0,1]$ be a parameter.

(1) Construct an initial triangulation $\mathcal{T}_0$ and set $l := 0$.
(2) *('Solve')* Compute the solution $u_l \in V(\mathcal{T}_l)$ of Problem (2.3).
(3) *('Estimate')* Compute $\eta_l(K)$, $K \in \mathcal{K}_l$, and η_l as well as $\delta_l(T)$, $T \in \mathcal{T}_l$, and δ_l.
(4) If $\eta_l + \delta_l = 0$, then STOP.
(5) *('Mark')* Determine a subset $\widehat{\mathcal{K}}_l$ of $\mathcal{K}_l$, such that the following conditions are fulfilled:

$$\sum_{K \in \widehat{\mathcal{K}}_l} \eta_l(K)^q \;\geq\; \theta^q \eta_l^q, \tag{2.9a}$$

$$\delta_l > \eta_l \Longrightarrow \sum_{T \in \widehat{\mathcal{K}}_l \cap \mathcal{T}_l} \delta_l(T)^q \;\geq\; \theta^q \delta_l^q \tag{2.9b}$$

(6) *('Refine')* Refine $\mathcal{T}_l$ by using bisection as to obtain $\mathcal{T}_{l+1}$ with

$$\forall\, K \in \widehat{\mathcal{K}}_l \quad \varphi_l^K \in V(\mathcal{T}_{l+1}). \tag{2.10}$$

(7) Increment l and go to step 2.

The stopping test in step (4) is justified by Remark 3.4 below. The use of bisection in step 'Refine' guarantees that, for all l,

$$V(\mathcal{T}_l) \subset V(\mathcal{T}_{l+1}) \tag{2.11}$$

and

$$\gamma(\mathcal{T}_l) \leq \gamma^*, \tag{2.12}$$

where γ^* depends only on the initial triangulation $\mathcal{T}_0$ and the dimension d; see e.g. Schmidt and Siebert [22] as well as the references therein. Of course, one can use other refinement techniques which satisfy the three conditions (2.10), (2.11), and (2.12).

*Remark 2.4 (*Test below tolerance) The stopping test in step (4) may be replaced by a test whether $\eta_l + \delta_l$ is below a given tolerance. However, this has efficiency problems and, for $p \in]1,2[$, it is not directly (i.e. up to a constant depending on d, p, and γ^*) related to the error in the energy norm; see Remark 3.5 below.

*Remark 2.5 (*Data oscillation marking) The marking criterion (2.9b) concerning data oscillation is different from [19,20]. It avoids overrefinement due to the reduction of data oscillation without any a priori knowledge of the overall convergence speed.

*Remark 2.6 (*Choice of $\widehat{\mathcal{K}}_l$) For efficiency reasons, one should try to choose $\widehat{\mathcal{K}}_l$ in step 'Mark' as small as possible. Consequently, it is convenient to collect in $\widehat{\mathcal{K}}_l$ those elements with relatively big indicators. The optimal construction of $\widehat{\mathcal{K}}_l$ requires a sort with complexity $\#\mathcal{K}_l \log \#\mathcal{K}_l$. Another possibility is the quasi-optimal construction proposed by Dörfler [13], which has complexity $\#\mathcal{K}_l$.

*Remark 2.7 (*Marking & Overrefinement) Taking into account the preceding remark and (2.8) yields the following: the regions where the mesh is refined by Algorithm 2.1 have a relatively big residual in the local dual energy norm.

3 Convergence

The purpose of this section is to establish the main result, which is stated in the introduction too. As already mentioned there, Algorithm 2.1 with $\theta < 1$ does not lead to $\max_{T\in\mathcal{T}_l} h_T \to 0$ as $l \to \infty$ necessarily. In other words: we prove convergence *without assuming* that $(V_l)_l$ approximates V in the sense $V = \overline{\bigcup_l V_l}$ with respect to the energy norm $\|\!|\cdot|\!\|_\Omega$.

Theorem 3.1 **(**Convergence) *Let $u \in V$ be the solution of Problem (2.1). Algorithm 2.1 converges in a finite number of iterations or produces an infinite sequence $(u_l)_l \subset V$ of approximate solutions with*

$$\|\!|u_l - u|\!\|_\Omega \to 0 \quad \textit{as} \quad l \to \infty.$$

The *proof of Theorem 3.1* is divided into several steps. The same letter C is used to denote different constants that depend only on d, p, and γ^* in (2.12).

We suppose that Algorithm 2.1 never stops and so produces an infinite sequence $(u_l)_l \subset V$ of approximate solutions; the case of a finite number of iterations will be treated along.

3.1 Monotonicity of energy

Our starting point is property (2.11). It implies that $\mathbb{N}_0 \ni l \mapsto J(u_l) \in \mathbb{R}$ is decreasing. More precisely, since Problems (2.1) and (2.2) as well as Problems (2.3) and (2.4) are equivalent, we have

$$J(u_l) \;\geq\; J(u_{l+1}) \;\geq\; J(u)$$

for any $l \in \mathbb{N}_0$. The sequence $(J(u_l))_l$ thus converges in $\mathbb{R}$. Therefore,

$$\lim_{l\to\infty} [J(u_l) - J(u_{l+1})] = 0, \tag{3.1}$$

i.e. the energy reduction of one adaptive iteration decreases to 0. If the adaption of such an iteration is efficient, this *should* imply

$$\lim_{l\to\infty} \|DJ(u_l)\|_{\Omega;*} = 0, \tag{3.2}$$

i.e. the residual of u_l tends to 0.

We shall refer to statement (3.2) as *consistency of Algorithm 2.1*. The main part of the rest of this section is devoted to the claimed relationship of (3.1) and (3.2).

3.2 Bounding the residual from above

The "decision making" in Algorithm 2.1 is done by means of *local* computable quantities. We have to relate them to the *global* quantities $\|DJ(u_l)\|_{\Omega;*}$ and $J(u_l) - J(u_{l+1})$. In this subsection we establish the relationship for the residual $\|DJ(u_l)\|_{\Omega;*}$ and discuss first consequences.

Lemma 3.1 **(**Upper bound for the residual) *There is a constant C depending only on d, p and γ^* in (2.12) such that*

$$\|DJ(u_l)\|_{\Omega;*} \le C\,(\eta_l + \delta_l) \tag{3.3}$$

for all $l \in \mathbb{N}_0$.

Proof. We consider only the more involved case $d \ge 2$; the proof for $d = 1$ goes along the same lines with the difference that there are no jump terms in steps 2 and 3, because one can use the Lagrange (instead of the Clément) interpolant in step 3.

1. We begin with a motivating observation. Piecewise integration by parts yields the following representation formula for the residual $DJ(u_l)$:

$$\begin{aligned}\langle DJ(u_l), \varphi\rangle = &- \sum_{S\in\mathcal{S}_l} \int_S [\![a(\nabla u_l)\cdot n]\!]\,\varphi - \sum_{T\in\mathcal{T}_l} \int_T f_T\varphi \\ &+ \sum_{T\in\mathcal{T}_l} \int_T (f_T - f)\varphi\end{aligned} \tag{3.4}$$

for all $\varphi \in V$, where $[\![a(\nabla u_l)\cdot n]\!]_{|S}$ denotes the jump in the normal component of $a(\nabla u_l)$ across $S \in \mathcal{S}_l$. This representation consists of two parts: the first two sums are "finite-dimensional" (in the sense that they are determined

by the finite real-valued sequences $(\llbracket a(\nabla u_l)\cdot n\rrbracket_{|S})_{S\in\mathcal{S}_l}$ and $(f_T)_{T\in\mathcal{T}_l}$), while the third sum is still "infinite-dimensional" (the functions $(f_T - f)_{|T}$, $T\in\mathcal{T}_l$, live generically in infinite-dimensional spaces). However, the third sum involves only the approximation of the data f. This suggests that $\|\!|DJ(u_l)|\!\|_{\Omega;*}$ can be controlled by a finite number of test functions and data oscillation. To prove this, we adapt ideas of Morin *et al.* [20, Section 3] to the context at hand.

2. The key tool to control the "finite-dimensional" part in (3.4) is the linear operator $\Pi_l : V \to V_l^+ := \operatorname{span}\{\varphi_l^K \mid K\in\mathcal{K}_l\}$ defined by

$$\forall\, K\in\mathcal{K}_l \quad \int_K \Pi_l\varphi = \int_K \varphi. \tag{3.5}$$

These $\#\mathcal{K}_l$ linear conditions determine a unique element of the $\#\mathcal{K}_l$-dimensional space V_l^+, because, if $\Pi_l\varphi = \sum_{K\in\mathcal{K}_l}\beta_l^K\varphi_l^K$, then $\int_K \Pi_l\varphi = 0$ for all $K\in\mathcal{K}_l$ implies $\beta_l^K = 0$ for all $K\in\mathcal{K}_l$. In fact, noting $\operatorname{supp}\varphi_l^K \subset \omega_l^K := \cup\{T\in\mathcal{T}_l : T\supset K\}$, we deduce first $\beta_l^S = 0$ for all $S\in\mathcal{S}_l$ and then $\beta_l^{\mathrm{T}} = 0$ for all $T\in\mathcal{T}_l$.

In what follows, we will need the following local stability-type estimates for the linear operator Π_l: if $\Pi_l\varphi = \sum_{K\in\mathcal{K}_l}\beta_l^K\varphi_l^K$, then

$$|\beta_l^S| \;\le\; C\, h_S^{-1/q}\,\|\varphi\|_{p;S}\,, \tag{3.6a}$$

$$|\beta_l^{\mathrm{T}}| \;\le\; C\left(h_T^{-1}\,\|\varphi\|_{p;T} + \sum_{\substack{S\in\mathcal{S}_l\\ \text{with}\,S\subset\partial T}} h_S^{-1/q}\,\|\varphi\|_{p;S}\right) \tag{3.6b}$$

for all $S\in\mathcal{S}_l$ and all $T\in\mathcal{T}_l$. These estimates follow from the Hölder inequality as well as $\int_S\varphi_l^S \ge Ch_S^{1/q}|S|^{1/q}$ and $\int_T\varphi_l^{\mathrm{T}} \ge Ch_T|T|^{1/q}$, which in turn are consequences of the normalization $\|\!|\varphi_l^K|\!\|_\Omega = 1$, $K\in\mathcal{K}_l$.

3. Let $\varphi\in V$ be a "continuous" test function and $\Lambda_l\varphi\in V_l$ be its Clément interpolant; see [10]. Using the definition of u_l, the representation formula (3.4), and the definition (3.5) of Π_l, we observe

$$\begin{aligned}\langle DJ(u_l),\varphi\rangle = \langle DJ(u_l),\varphi-\Lambda_l\varphi\rangle \;&=\; \sum_{K\in\mathcal{K}_l}\beta_l^K\langle DJ(u_l),\varphi_l^K\rangle \\ &\quad - \sum_{T\in\mathcal{T}_l}\int_T (f_T - f)[\Pi_l(\varphi-\Lambda_l\varphi) - (\varphi-\Lambda_l\varphi)],\end{aligned} \tag{3.7}$$

where β_l^K, $K\in\mathcal{K}_l$, is defined by $\Pi_l(\varphi-\Lambda_l\varphi) = \sum_{K\in\mathcal{K}_l}\beta_l^K\varphi_l^K$.

We estimate the first term of the right-hand side in (3.7). We derive

$$|\beta_l^S| \;\le\; C\, h_S^{-1/q}\,\|\Lambda_l\varphi-\varphi\|_{p;S} \;\le\; C\,\|\!|\varphi|\!\|_{\tilde\omega_l^S}\,, \tag{3.8a}$$

$$(3.8b)\quad |\beta_l^{\mathrm{T}}| \le C \left(h_T^{-1} \|\Lambda_l\varphi - \varphi\|_{p;T} + \sum_{\substack{S\in\mathcal{S}_l \\ \text{with } S\subset\partial T}} h_S^{-1/q} \|\Lambda_l\varphi - \varphi\|_{p;S} \right) \le C \, |||\varphi|||_{\tilde{\omega}_l^{\mathrm{T}}} ,$$

with $\tilde{\omega}_l^K := \{T \in \mathcal{T}_l \mid T \cap K \neq \emptyset\}$, $K \in \mathcal{K}_l$, by exploiting (3.6), consequences of (2.12) on the local mesh, and local approximation properties of Λ_l; for the latter ones, see e.g. Verfürth [23, Lemma 3.1] and Clément [10]. Thus, we arrive at

$$(3.9)\quad \sum_{K\in\mathcal{K}_l} \beta_l^K \langle DJ(u_l), \varphi_l^K\rangle \;\le\; \sum_{K\in\mathcal{K}_l} |\beta_l^K| \, \eta_l(K) \;\le\; C\eta_l \, |||\varphi|||_\Omega$$

with the help of $\#\{\tilde{T} \in \mathcal{T}; \tilde{T} \cap T \neq \emptyset\} \le C$ for all $T \in \mathcal{T}_l$.

It remains to estimate the second sum of the right-hand side in (3.7) appropriately. To this end, we first observe

$$\|\Pi_l(\Lambda_l\varphi - \varphi)\|_{p;T} \;\le\; \sum_{\substack{K\in\mathcal{K}_l \\ \text{with} K\subset T}} |\beta_l^K| \, \left\|\varphi_l^K\right\|_{p;T} \;\le\; C\, h_T \, |||\varphi|||_{\tilde{\omega}_l^{\mathrm{T}}}$$

by $\left\|\varphi_l^K\right\|_{p;T} \le Ch_T$ and (3.8). Consequently,

$$(3.10)\qquad -\sum_{T\in\mathcal{T}_l} \int_T (f_T - f)[\Pi_l(\varphi - \Lambda_l\varphi) - (\varphi - \Lambda_l\varphi)] \le \sum_{T\in\mathcal{K}_l} \delta_l(T) \, |||\varphi|||_{\tilde{\omega}_l^{\mathrm{T}}} \le \delta_l \, |||\varphi|||_\Omega .$$

*Remark 3.1 (*Without standard residual indicators) In contrast to the proofs of [23, Propositions 1.14 and 1.15], we do not employ the standard residual indicators (see Section 3.6.3 below) in establishing (3.3).

Lemma 3.1 suggests to show (3.2) by establishing

$$\lim_{l\to\infty} \eta_l = 0 \quad \text{and} \quad \lim_{l\to\infty} \delta_l = 0.$$

The first limit is the crucial one. This can be seen from the following observation.

Lemma 3.2 (Reduction of data oscillation) *Suppose that* $\lim_{l\to\infty} \eta_l = 0$. *Then* $\lim_{l\to\infty} \delta_l = 0$.

Proof. We first observe that (2.11), (2.9b), and (2.10) imply

$$\forall\, l \in \mathbb{N}_0 \quad \delta_{l+1} \leq \delta_l, \tag{3.11a}$$

$$\exists\, \alpha \in]0,1[\ \ \forall\, l \in \mathbb{N}_0 \quad \delta_l > \eta_l \Longrightarrow \delta_{l+1} \ \leq\ \alpha \delta_l; \tag{3.11b}$$

see [19, Lemma 3.8]. Next, we consider two cases:

Case 1: $\forall\, l \in \mathbb{N}_0 \ \ \exists\, m > l \quad \delta_m \leq \eta_m$. Then the claim follows from the assumption $\lim_{l\to\infty} \eta_l = 0$ and the monotonicity (3.11a) of $(\delta_l)_l$.

Case 2: $\exists\, l \in \mathbb{N}_0 \ \ \forall\, m \geq l \quad \delta_m > \eta_m$. Here, the claim follows from the strict reduction of the data oscillation (3.11b).

3.3 Bounding the energy reduction from below

In order to relate the computable quantities in Algorithm 2.1 to the energy reduction $J(u_l) - J(u_{l+1})$, we introduce the following *local* and discrete variants of Problem (2.2).

Problem 3.1 (Variational formulation in a test direction) *Let* $l \in \mathbb{N}_0$ *and* $K \in \mathcal{K}_l$. *Find* $\alpha_l(K) \in \mathbb{R}$ *such that*

$$\forall\, \beta \in \mathbb{R} \quad J\left(u_l + \alpha_l(K)\, \varphi_l^K\right) \ \leq\ J\left(u_l + \beta\, \varphi_l^K\right).$$

Problem (3.1) is uniquely solvable and, thus, $\alpha_l(K)$ is well defined. Note that

$$\int_\Omega a\left(\nabla\left[u_l + \alpha_l(K)\, \varphi_l^K\right]\right) \cdot \nabla \varphi_l^K \ =\ \int_\Omega f \varphi_l^K. \tag{3.12}$$

For any $l \in \mathbb{N}_0$ and $K \in \mathcal{K}_l$, we define

$$\zeta_l(K) := J(u_l) - J\left(u_l + \alpha_l(K)\, \varphi_l^K\right).$$

This number $\zeta_l(K)$ is a lower bound for the energy reduction that is achieved, if the direction φ_l^K is added to the current finite element space $V(\mathcal{T}_l)$. The number $|\alpha_l(K)|$ expresses this lower bound as the energy norm of the corresponding correction.

The following lemma relates the energy reduction $J(u_l) - J(u_{l+1})$ to certain local minimizations.

Lemma 3.3 (Lower bound for the energy reduction) *For all* $l \in \mathbb{N}_0$, *we have*

$$J(u_l) - J(u_{l+1}) \ \geq\ \frac{\widehat{\zeta}_l}{d+2}$$

with $\widehat{\zeta}_l := \sum_{K\in\widehat{\mathcal{K}}_l} \zeta_l(K)$ *and* $\widehat{\mathcal{K}}_l$ *as in step 'Mark'.*

Remark 3.2 Note that the lower bound in Lemma 3.3 involves neither p nor γ^* of (2.12).

Proof of Lemma 3.3. It is convenient to introduce the following notations and quantities. Let $\overline{\mathcal{K}}_l$ be the union of $\mathcal{K}_l$ and the set of all boundary sides. We define

$$\widehat{\zeta}_l(K) := \begin{cases} \zeta_l(K), \text{ if } K \in \widehat{\mathcal{K}}_l; \\ 0, \qquad \text{otherwise;} \end{cases} \qquad \widehat{\alpha}_l(K) := \begin{cases} \alpha_l(K), \text{ if } K \in \widehat{\mathcal{K}}_l; \\ 0, \qquad \text{otherwise;} \end{cases}$$

for all $K \in \overline{\mathcal{K}}_l$ and set

$$\widehat{\varphi} \;:=\; \frac{1}{d+2} \sum_{K \in \overline{\mathcal{K}}_l} \widehat{\alpha}_l(K) \varphi_l^K.$$

Because of (2.10) we have $\widehat{\varphi} \in V(\mathcal{T}_{l+1})$ and thus

$$J(u_{l+1}) \;\le\; J(u_l + \widehat{\varphi}) \;=\; \sum_{T \in \mathcal{T}_l} \int_T \left[\frac{1}{p} |\nabla(u_l + \widehat{\varphi})|^p - f(u_l + \widehat{\varphi}) \right].$$

We further observe that, for any $T \in \mathcal{T}_l$,

$$(u_l + \widehat{\varphi})_{|T} \;=\; \sum_{K \in \mathcal{K}_l(T)} \frac{v_{K|T}}{d+2},$$

where $\mathcal{K}_l(T) := \{K \in \overline{\mathcal{K}}_l \mid K \subset T\}$ and $v_K := u_l + \widehat{\alpha}_l(K)\, \varphi_l^K$ (we suppress the dependence on l in the notation). The convexity of the function $\mathbb{R}^d \ni \xi \mapsto |\xi|^p \in \mathbb{R}$ thus implies

$$\begin{aligned} \int_T \left[\frac{1}{p} |\nabla(u_l + \widehat{\varphi})|^p - f(u_l + \widehat{\varphi}) \right] \\ \le \frac{1}{d+2} \sum_{K \in \mathcal{K}_l(T)} \int_T \left(\frac{1}{p} |\nabla v_K|^p - f v_K \right), \end{aligned}$$

whence

$$\begin{aligned} J(u_{l+1}) &\le \frac{1}{d+2} \sum_{T \in \mathcal{T}} \sum_{K \in \mathcal{K}_l(T)} \int_T \left(\frac{1}{p} |\nabla v_K|^p - f v_K \right), \\ &= \frac{1}{d+2} \sum_{K \in \overline{\mathcal{K}}_l} \int_{\omega_l^K} \left(\frac{1}{p} |\nabla v_K|^p - f v_K \right). \end{aligned}$$

Hereafter, $\omega_l^K := \bigcup\{T \in \mathcal{T}_l \,:\, T \supset K\}$. Since

$$\int_{\omega_l^K}\left(\frac{1}{p}|\nabla v_K|^p - f v_K\right) - \int_{\omega_l^K}\left(\frac{1}{p}|\nabla u_l|^p - f u_l\right) = J\left(u_l + \widehat{\alpha}_l(K)\,\varphi_l^K\right) - J(u_l) \;=\; -\widehat{\zeta}_l(K)$$

for any $K \in \overline{\mathcal{K}}_l$, we conclude

$$\begin{aligned} J(u_{l+1}) &\le \frac{1}{d+2}\sum_{K\in\overline{\mathcal{K}}_l}\left[\int_{\omega_l^K}\left(\frac{1}{p}|\nabla u_l|^p - f u_l\right) - \widehat{\zeta}_l(K)\right] \\ &= J(u_l) \;-\; \frac{\widehat{\zeta}_l}{d+2}. \end{aligned}$$

□

3.4 Local minimization and local error indicator

On the one hand, we have bounded the residual $\|\!|DJ(u_l)|\!\|_{\Omega;*}$ in terms of the local error indicators $\eta_l(K)$, $K \in \mathcal{K}_l$, and data oscillation. On the other hand, we have bounded the local energy reduction indicators $\zeta_l(K), K \in \widehat{\mathcal{K}}_l$, in terms of the global energy reduction $J(u_l) - J(u_{l+1})$. Therefore, in order to relate $\|\!|DJ(u_l)|\!\|_{\Omega;*}$ and $J(u_l) - J(u_{l+1})$, we still miss a bound of $\eta_l(K)$ in terms of $\zeta_l(K)$, $K \in \mathcal{K}_l$. In this subsection we analyze the relationship between $\zeta_l(K)$, $\eta_l(K)$, and $\alpha_l(K)$, $K \in \mathcal{K}_l$. In particular, we provide the missing bound.

Lemma 3.4 *For $l \in \mathbb{N}_0$ and $K \in \mathcal{K}_l$, there holds*

$$C\,|\alpha_l(K)|\,\eta_l(K) \;\le\; \zeta_l(K) \;\le\; |\alpha_l(K)|\,\eta_l(K), \tag{3.13}$$

where

$$C \;\ge\; \begin{cases} 2^{p-3}\,\dfrac{C_2}{C_1}, & \text{if } p \in]1,2], \\ \dfrac{C_2}{p\,C_1}, & \text{if } p \in [2,\infty[, \end{cases}$$

and C_1, C_2 as in (2.1).

Proof. It is convenient to define $v_K := u_l + \alpha_l(K)\varphi_l^K$.

1. We first show the second inequality in (3.13). We have

$$-\zeta_l(K) \;=\; J(v_K) - J(u_l) \;\ge\; \alpha_l(K)\,\left\langle DJ(u_l), \varphi_l^K\right\rangle$$

thanks to the convexity of J. Since $\zeta_l(K) \geq 0$ and $\eta_l(K) = |\langle DJ(u_l), \varphi_l^K \rangle|$, we obtain

$$\zeta_l(K) \;\leq\; |\alpha_l(K)|\, \eta_l(K).$$

2. In order to prove the first inequality in (3.13), we start with an upper bound of $\eta_l(K)$. Identity (3.12) yields

$$\eta_l(K) = \left| \int_\Omega [a(\nabla u_l) - a(\nabla v_K)] \cdot \nabla \varphi_l^K \right|.$$

Therefore,

$$\eta_l(K) \;\leq\; C_1\, |\alpha_l(K)| \int_\Omega (|\nabla u_l| + |\nabla v_K|)^{p-2} |\nabla \varphi_l^K|^2 \tag{3.14}$$

due to the continuity property (2.1a).

3. Next, we provide a lower bound for $\zeta_l(K)$. Making use of identity (3.12) a second time, we obtain the formula

$$\begin{aligned} \zeta_l(K) &= \int_0^1 \langle DJ(su_l + (1-s)v_K) - DJ(v_K),\, u_l - v_K \rangle \; ds \\ &= \int_0^1 \int_\Omega \{a(\nabla[v_K + s(u_l - v_K)]) - a(\nabla v_K)\} \\ &\qquad \cdot \nabla\{[v_K + s(u_l - v_K)] - v_K\} \, \frac{ds}{s}. \end{aligned}$$

Thus, the ellipticity property (2.1b) leads to

$$\begin{aligned} &\zeta_l(K) \\ &\geq C_2 |\alpha_l(K)|^2 \int_0^1 s \int_\Omega (|\nabla[v_K + s(u_l - v_K)]| + |\nabla v_K|)^{p-2} |\nabla \varphi_l^K|^2 ds. \end{aligned} \tag{3.15}$$

4. In order to conclude, we have to relate the terms $(|\nabla u_l| + |\nabla v_K|)^{p-2}$ and $(|\nabla[v_K + s(u_l - v_K)]| + |\nabla v_K|)^{p-2}$. To this end, we consider the two cases $p \in]1, 2]$ and $p \geq 2$ separately.

Case 1: $-1 < p - 2 \leq 0$. Then

$$|\nabla[v_K + s(u_l - v_K)]| + |\nabla v_K| \;\leq\; 2\,(|\nabla u_l| + |\nabla v_K|)$$

implies

$$\begin{aligned} &\int_\Omega (|\nabla u_l| + |\nabla v_K|)^{p-2} |\nabla \varphi_l^K|^2 \\ &\leq\; 2^{3-p} \int_0^1 s \int_\Omega (|\nabla[v_K + s(u_l - v_K)]| + |\nabla v_K|)^{p-2} \, |\nabla \varphi_l^K|^2 \, ds. \end{aligned}$$

Consequently, the first inequality in (3.13) holds with $C \geq 2^{p-3} C_2/C_1$, if $p \in]1,2]$.

Case 2: $p-2 \geq 0$. Here,

$$|\nabla[v_K + s(u_l - v_K)]| + |\nabla v_K| \;\geq\; s\,(|\nabla u_l| + |\nabla v_K|) \quad \text{for all } s \in [0,1]$$

implies

$$\begin{aligned}&\int_\Omega (|\nabla u_l| + |\nabla v_K|)^{p-2} |\nabla \varphi_l^K|^2 \\ &\leq\; p \int_0^1 s \int_\Omega (|\nabla[v_K + s(u_l - v_K)]| + |\nabla v_K|)^{p-2}\, |\nabla \varphi_l^K|^2 \, ds.\end{aligned}$$

Therefore, the first inequality in (3.13) holds with $C \geq C_2/(pC_1)$, if $p \geq 2$.

Remark 3.3 The calculations of steps 3 and 4 in the preceding proof can be found in the proof of [4, Theorem 2.1], too.

Lemma 3.5 *There is a constant C depending only on d and p such that, for all $l \in \mathbb{N}_0$ and $K \in \mathcal{K}_l$, there holds:*

- *If $p \in]1,2]$, then*

$$\frac{1}{C}\,\eta_l(K)^q \;\leq\; \zeta_l(K) \;\leq\; C\,\eta_l(K)^2 \left(\|\!| u_l |\!\|_{\operatorname{supp} \varphi_l^K} + \zeta_l(K)^{1/p} \right)^{2-p}.$$

- *If $p \geq 2$, then*

$$\frac{1}{C}\,\zeta_l(K) \;\leq\; \eta_l(K)^q \;\leq\; C\,\zeta_l(K)^{q/2} \left(\|\!| u_l |\!\|_{\operatorname{supp} \varphi_l^K} + \zeta_l(K)^{1/p} \right)^{\frac{p-2}{2} q}.$$

Proof. **1.** We establish the first inequality in the case $p \in]1,2]$. Inserting

$$|\nabla u_l| + |\nabla v_K| \;\geq\; |\nabla(u_l - v_K)| \;=\; |\alpha_l(K)|\,|\nabla \varphi_l^K|$$

in (3.14) and taking into account $p-2 \leq 0$ as well as $\left\| \varphi_l^K \right\|_\Omega = 1$ yields

$$\eta_l(K) \;\leq\; C_1\,|\alpha_l(K)|^{p-1}. \tag{3.16}$$

This, in conjunction with the first inequality in Lemma 3.4, leads to

$$C_1^{\frac{-1}{p-1}} C\,\eta_l(K)^q \;=\; C_1^{\frac{-1}{p-1}} C\,\eta_l(K)^{\frac{-1}{p-1}+1} \;\leq\; C\,|\alpha_l(K)|\,\eta_l(K) \;\leq\; \zeta_l(K).$$

2. We prove the second inequality in the case $p \in]1,2]$. To this end, we first show

$$|\alpha_l(K)|^2 \;\leq\; C\,\zeta_l(K) \left(\|\!| u_l |\!\|_{\operatorname{supp} \varphi_l^K} + |\alpha_l(K)| \right)^{2-p}. \tag{3.17}$$

Without loss of generality we may assume that $\alpha_l(K) \neq 0$. Following the proof of [4, Lemma 2.2], we have

$$
\begin{aligned}
(3.18)\quad & |\alpha_l(K)|^2 \\
&\leq \left(\int_\Omega \left(|\nabla u_l| + |\nabla[\alpha_l(K)\varphi_l^K]| \right)^{(p-2)\left(-\frac{p}{2}+\frac{p}{2}\right)} |\nabla[\alpha_l(K)\varphi_l^K]|^p \right)^{2/p} \\
&\leq \left(\int_{\operatorname{supp}\varphi_l^K} \left(|\nabla u_l| + |\nabla[\alpha_l(K)\varphi_l^K]| \right)^p \right)^{(2-p)/p} \\
&\qquad \int_\Omega \left(|\nabla u_l| + |\nabla[\alpha_l(K)\varphi_l^K]| \right)^{p-2} |\nabla[\alpha_l(K)\varphi_l^K]|^2 \\
&\leq \left(\|u_l\|_{\operatorname{supp}\varphi_l^K} + |\alpha_l(K)| \right)^{2-p} \\
&\qquad |\alpha_l(K)|^2 \int_\Omega \left(|\nabla u_l| + |\nabla[\alpha_l(K)\varphi_l^K]| \right)^{p-2} |\nabla\varphi_l^K|^2
\end{aligned}
$$

by applying $\left\|\varphi_l^K\right\|_\Omega = 1$, a Hölder inequality, and the triangle inequality. Since

$$
\left(|\nabla u_l| + |\nabla[\alpha_l(K)\varphi_l^K]| \right)^{p-2} \;\leq\; 2^{2-p} (|\nabla[v_K + s(u_l - v_K)]| + |\nabla v_K|)^{p-2}
$$

thanks to $p - 2 \leq 0$, we obtain (3.17) with the help of (3.15).

Next, we show that

$$
(3.19)\quad t \geq 0, \quad t^2 \;\leq\; B\,(A+t)^{2-p} \quad\Longrightarrow\quad t \;\leq\; C\,\max\{B^{1/p}, A\}
$$

for A, $B \geq 0$. So, suppose that the premise of (3.19) is satisfied. Since for every $\rho \geq 0$ there is a constant C_ρ with

$$
(3.20)\qquad \forall\, s, t \in \mathbb{R} \quad (s+t)^\rho \;\leq\; C_\rho\,(s^\rho + t^\rho)\,,
$$

we obtain

$$
t^2 \;\leq\; B\,(A+t)^{2-p} \;\leq\; C\,\left(BA^{2-p} + Bt^{2-p}\right),
$$

whence

$$
(3.21)\qquad t^{2-p}\,(t^p - CB) \;\leq\; CBA^{2-p}.
$$

Let us consider two cases.

Case 1: $t^p - CB \leq B$. Then $t^p \leq (C+1)\,B$ and hence $t \leq C\,B^{1/p}$.

Case 2: $t^p - CB > B$. Since $B \geq 0$, we derive

$$
t^{2-p} \;\leq\; C\,\frac{BA^{2-p}}{t^p - CB} \;<\; C\,A^{2-p}
$$

from (3.21). We thus obtain $t \leq CA$ as $p \in]1,2[$ (the case $p = 2$ is trivial).

Taking into account both cases, we have established (3.19).

The inequalities (3.17) and (3.19) imply

$$|\alpha_l(K)|^2 \leq C\,\zeta_l(K)\left(\|u_l\|_{\operatorname{supp}\varphi_l^K} + \zeta_l(K)^{1/p}\right)^{2-p}. \tag{3.22}$$

In view of the second inequality of Lemma (3.4), we finally arrive at

$$\begin{aligned}\zeta_l(k) &\leq |\alpha_l(K)|\,\eta_l(K)\\ &\leq C\eta_l(K)\zeta_l(K)^{1/2}\left(\|u_l\|_{\operatorname{supp}\varphi_l^K} + \zeta_l(K)^{1/p}\right)^{(2-p)/2}.\end{aligned}$$

This completes the proof for the case $p \in]1,2]$.

3. We show the first inequality in the case $p \geq 2$. Since

$$|\nabla[v_K + s(u_l - v_K)]| + |\nabla v_K| \geq s\,|\nabla[\alpha_l(K)\varphi_l^K]| \quad \text{for all } s \in [0,1]$$

and $p - 2 \geq 0$, we derive from (3.15) the inequality

$$\zeta_l(K) \geq \frac{C_2}{p}\,|\alpha_l(K)|^p. \tag{3.23}$$

Inserting this into the second inequality of Lemma 3.4 yields

$$\zeta_l(K) \leq |\alpha_l(K)|\,\eta_l(K) \leq \left(\frac{p}{C_2}\right)^{1/p}\zeta_l(K)^{1/p}\eta_l(K),$$

which implies the first inequality for $p \geq 2$.

4. Finally, we establish the second inequality in the case $p \geq 2$. To this end, we first prove

$$\eta_l(K) \leq C\,|\alpha_l(K)|\left(\|u_l\|_{\operatorname{supp}\varphi_l^K} + |\alpha_l(K)|\right)^{p-2}. \tag{3.24}$$

Using

$$|\nabla u_l| + |\nabla v_K| \leq 2\left(|\nabla u_l| + |\nabla[\alpha_l(K)\varphi_l^K]|\right),$$

$p - 2 \geq 0$, a Hölder inequality, and $\left\|\varphi_l^K\right\|_\Omega = 1$ in (3.14), we obtain (without loss of generality $p > 2$)

$$\begin{aligned}\eta_l(K) &\leq 2^{p-2}\,C_1\,|\alpha_l(K)|\int_\Omega\left(|\nabla u_l| + |\nabla[\alpha_l(K)\varphi_l^K]|\right)^{p-2}|\nabla\varphi_l^K|^2\\ &\leq 2^{p-2}\,C_1\,|\alpha_l(K)|\left(\int_\Omega\left(|\nabla u_l| + |\nabla[\alpha_l(K)\varphi_l^K]|\right)^p\right)^{(p-2)/p}\\ &\leq 2^{p-2}\,C_1\,|\alpha_l(K)|\left(\|u_l\|_{\operatorname{supp}\varphi_l^K} + |\alpha_l(K)|\right)^{p-2},\end{aligned}$$

whence (3.24). We insert (3.23) for the second occurrence of $|\alpha_l(K)|$ in (3.24) and get

$$\eta_l(K) \;\le\; C\,|\alpha_l(K)|\,\Big(\|\!|u_l|\!\|_{\operatorname{supp}\varphi_l^K} \,+\, \zeta_l(K)^{1/p}\Big)^{p-2}$$

Without loss of generality we may assume $\eta_l(K) > 0$. Accordingly, $|\alpha_l(K)| > 0$ and $\zeta_l(K) > 0$. Combing the last inequality with the first one in Lemma 3.4, we arrive at

$$\frac{\eta_l(K)^2}{(\|\!|u_l|\!\|_{\operatorname{supp}\varphi_l^K} + \zeta_l(K)^{1/p})^{(p-2)}} \;\le\; C\,|\alpha_l(K)|\,\eta_l(K) \;\le\; C\,\zeta_l(K)$$

This completes the proof of the second inequality for $p \ge 2$.

3.5 Consistency and convergence

The preceding subsections allow us to prove the following key estimates.

Proposition 3.1 (Control of Error Indicators) *There exists a constant C depending only on d and p such that, for all $l \in \mathbb{N}_0$:*

- *If $p \in]1, 2]$, then*

$$\eta_l \;\le\; \frac{C}{\theta}\,[J(u_l) - J(u_{l+1})]^{1/q}.$$

- *If $p \ge 2$, then*

$$\eta_l \;\le\; \frac{C}{\theta}[J(u_l) - J(u_{l+1})]^{1/2}\,\Big[\|\!|f|\!\|_{\Omega;*}^q + J(u_l) - J(u_{l+1})\Big]^{(p-2)/(2p)}.$$

Proof. **1.** We first deal with the case $p \in]1, 2]$. Making use of the marking condition (2.9a), Lemma 3.5, and Lemma 3.3, we obtain

$$\eta_l^q \;\le\; \theta^{-q} \sum_{K\in\widehat{\mathcal{K}}_l} \eta_l(K)^q \le C\theta^{-q} \sum_{K\in\widehat{\mathcal{K}}_l} \zeta_l(K) \le C\theta^{-q}[J(u_l) - J(u_{l+1})].$$

2. We now prove the estimate for the case $p \ge 2$. We derive

$$\begin{aligned}
\eta_l^q &\le C\,\theta^{-q} \sum_{K\in\widehat{\mathcal{K}}_l} \zeta_l(K)^{q/2}\,\Big(\|\!|u_l|\!\|_{\operatorname{supp}\varphi_l^K} + \zeta_l(K)^{1/p}\Big)^{(p-2)\frac{q}{2}} \\
&\le C\,\theta^{-q}\,\widehat{\zeta}_l^{q/2}\,\left(\sum_{K\in\widehat{\mathcal{K}}_l}\Big[\|\!|u_l|\!\|_{\operatorname{supp}\varphi_l^K} + \zeta_l(K)^{1/p}\Big]^p\right)^{\frac{p-2}{2}\frac{1}{p-1}} \\
&\le C\,\theta^{-q}\,[J(u_l) - J(u_{l+1})]^{q/2}\,\Big[\|\!|f|\!\|_{\Omega;*}^q + J(u_l) - J(u_{l+1})\Big]^{\frac{p-2}{2p}q}
\end{aligned}$$

by marking condition (2.9a), Lemma 3.5, a Hölder inequality, Lemma 3.3, (3.20) and stability estimate (2.5).

Proposition 3.1 has the following important consequence.

Corollary 3.1 (Consistency) *Algorithm 2.1 is consistent, i.e.*

$$\lim_{l\to\infty} \|DJ(u_l)\|_{\Omega;*} = 0.$$

Proof. The claim follows from Proposition 3.1, the convergence of the energy reduction (3.1), as well as Lemmata 3.2 and 3.1.

Combining Corollary 3.1 with the ellipticity properties in the energy norm of DJ completes the proof of Theorem 3.1.

We start with the uniformly monotone case $p \geq 2$. Inequality (2.3) implies

$$\begin{aligned} C_2 \, \|u_l - u\|_\Omega^p &\leq \int_\Omega [a(\nabla u_l) - a(\nabla u)] \cdot \nabla(u_l - u) \;=\; \langle DJ(u_l), u_l - u\rangle \\ &\leq \|DJ(u_l)\|_{\Omega;*} \, \|u_l - u\|_\Omega \,, \end{aligned}$$

whence

$$\|u_l - u\|_\Omega^{p-1} \;\leq\; C_2^{-1} \, \|DJ(u_l)\|_{\Omega;*} \,. \tag{3.25}$$

Consequently, Corollary 3.1 ensures

$$\lim_{l\to\infty} \|u_l - u\|_\Omega = 0$$

and Theorem 3.1 is established for $p \geq 2$.

It remains to deal with the non-uniformly monotone case $p \in]1, 2[$. A calculation similar to (3.18) yields

$$\|u_l - u\|_\Omega^2 \;\leq\; (\|u_l\|_\Omega + \|u\|_\Omega)^{2-p} \int_\Omega (|\nabla u_l| + |\nabla u|)^{p-2} |\nabla(u_l - u)|^2.$$

This, in conjunction with (2.1b), gives

$$\|u_l - u\|_\Omega \;\leq\; C_2^{-1} \, (\, \|u_l\|_\Omega + \|u\|_\Omega \,)^{2-p} \, \|DJ(u_l)\|_{\Omega;*} \tag{3.26}$$

(see also [16, Proposition 5.2]). Since

$$\|u_l\|_\Omega + \|u\|_\Omega \;\leq\; 2 \, \|f\|_{\Omega;*}^{1/(p-1)}$$

thanks to (2.4) and (2.5), we conclude, by Corollary 3.1,

$$\lim_{l\to\infty} \|u_l - u\|_\Omega = 0$$

also for $p \in]1, 2[$. The proof of Theorem 3.1 is thus completed, if Algorithm 2.1 never stops. The remaining case is treated in the following remark.

Remark 3.4 (Stopping test) Lemma 3.1 and, depending on p, (3.25) or (3.26) ensure that, if Algorithm 2.1 exits in step (4), then $u_l = u$.

*Remark 3.5 (*Gap between ellipticity and continuity) Complementing the ellipticity properties (3.25) and (3.26) with the continuity properties *in the energy norm*, we obtain the following: if $p \geq 2$, then

$$C_2 \, |||u_l - u|||_\Omega^{p-1} \;\leq\; |||DJ(u_l)|||_{\Omega;*} \;\leq\; C_1 (\, |||u_l|||_\Omega + |||u|||_\Omega \,)^{p-2} \, |||u_l - u|||_\Omega$$

(for the second inequality, use the Hölder inequality first with p and then with $p-1$) and, if $p \leq 2$, then

$$\begin{aligned} C_2 \, |||u_l - u|||_\Omega &\leq (\, |||u_l|||_\Omega + |||u|||_\Omega \,)^{2-p} \, |||DJ(u_l)|||_{\Omega;*} \\ &\leq C_1 (\, |||u_l|||_\Omega + |||u|||_\Omega \,)^{2-p} \, |||u_l - u|||_\Omega^{p-1} \, . \end{aligned}$$

We observe that in both inequalities the left-hand side vanishes asymptotically faster that the right-hand side as $|||u_l - u|||_\Omega \to 0$, if $p \neq 2$. This gap, which mimics the one between upper and lower bound in Verfürth [23, Proposition 3.12], entails that a stopping test below tolerance relying on $|||DJ(u_l)|||_{\Omega;*}$ has the following drawback: the algorithm may continue for quite a while although the error is already below tolerance. Moreover, the afore-mentioned gap prevents us from showing a convergence rate as in Morin *et al.* [19,20].

3.6 Other local error indicators

In this subsection we discuss several variants of Algorithm 2.1 that use different local error indicators. We mention only the modifications concerning the marking and refining conditions (2.9) and (2.10).

3.6.1 Local energy reduction indicators. One may replace the local error indicators $\eta_l(K)$, $K \in \mathcal{K}_l$, and η_l by

$$\zeta_l(K), \; K \in \mathcal{K}_l, \text{ and } \zeta_l := \sum_{K \in \mathcal{K}_l} \zeta_l(K).$$

Doing so, we substitute

$$\sum_{K \in \widehat{\mathcal{K}}_l} \zeta_l(K) \;\geq\; \theta \zeta_l \tag{3.27a}$$

$$\delta_l > \zeta_l^{1/q} \Longrightarrow \sum_{T \in \widehat{\mathcal{K}}_l \cap \mathcal{T}_l} \delta_l(T)^q \;\geq\; \theta^q \delta_l^q \tag{3.27b}$$

for the marking conditions (2.9). The modified algorithm still converges. In fact, (3.1), Lemma 3.3 and (3.27a) imply $\lim_{l\to\infty} \zeta_l = 0$. Slight modifications of Lemma 3.2 and Proposition 3.1 yield $\lim_{l\to\infty} \delta_l = 0$ and $\lim_{l\to\infty} \eta_l = 0$, respectively. The modified algorithm is thus also consistent and the convergence follows as before.

3.6.2 Local correction indicators. Another possibility might be to use

$$|\alpha_l(K)|,\ K \in \mathcal{K}_l, \text{ and } \alpha_l := \left(\sum_{K \in \mathcal{K}_l} |\alpha_l(K)|^p \right)^{1/p}$$

instead of $\eta_l(K)$, $K \in \mathcal{K}_l$, and η_l. In this case, (2.9) is replaced by

$$\sum_{K \in \widehat{\mathcal{K}}_l} |\alpha_l(K)|^p \ \geq\ \theta^p \alpha_l^p \tag{3.28a}$$

$$\delta_l > \alpha_l^{p-1} \Longrightarrow \sum_{T \in \widehat{\mathcal{K}}_l \cap \mathcal{T}_l} \delta_l(T)^q \ \geq\ \theta^q \delta_l^q. \tag{3.28b}$$

For the proof of convergence, we first observe that, depending on the value of p, (3.22) or (3.23) imply

$$\alpha_l \ \leq\ C\theta^{-1}[J(u_l) - J(u_{l+1})]^{1/2} \left[|\!|\!| f |\!|\!|_{\Omega;*}^q + J(u_l) - J(u_{l+1}) \right]^{(2-p)/(2p)}$$

or

$$\alpha_l \ \leq\ C\theta^{-1}[J(u_l) - J(u_{l+1})]^{1/p},$$

respectively. Consequently, one obtains $\lim_{l\to\infty} \alpha_l = 0$ and $\lim_{l\to\infty} \delta_l = 0$. Moreover, again depending on p, (3.16) or (3.24) yield

$$\eta_l \ \leq\ C_1^{1/q} \alpha_l^{p-1}$$

or

$$\eta_l \ \leq\ C\alpha_l \left(|\!|\!| f |\!|\!|_{\Omega;*}^q + \alpha_l^p \right)^{(p-2)/p},$$

respectively. Therefore, $\lim_{l\to\infty} \eta_l = 0$ and this variant of Algorithm 2.1 is consistent and converges, too.

3.6.3 Standard residual indicators. In [3, Section 2] the estimator

$$\nu_l := \left(\sum_{T \in \mathcal{T}_l} \nu_l(T)^q \right)^{1/q}$$

with

$$\nu_l(T) := \left(h_T \, \|[\![a(\nabla u_l) \cdot n]\!]\|_{q;\partial T \setminus \partial\Omega}^q \ +\ h_T^q \, \|f\|_{q;T}^q \right)^{1/q}, \quad T \in \mathcal{T}_l,$$

is proposed (we restrict ourselves to the more interesting case $d \geq 2$). Note that these indicators are indexed only by simplices. Accordingly, we construct in step 'Mark' a subset $\widehat{\mathcal{T}}_l$ of $\mathcal{T}_l$ rather than a subset of $\mathcal{K}_l$. Moreover,

$\nu_l(T)$ incorporates $\delta_l(T)$ partially and we have to deal with data oscillation separately. We thus substitute

$$\sum_{T\in\widehat{\mathcal{T}}_l} \nu_l(T)^q \;\geq\; \theta^q \nu_l^q \tag{3.29a}$$

$$\delta_l > \delta_0\rho^l \Longrightarrow \sum_{T\in\widehat{\mathcal{T}}_l} \delta_l(T)^q \;\geq\; \theta^q\delta_l^q\theta^q\delta_l^q \tag{3.29b}$$

for (2.9). Here, $\rho\in\,]0,1[$ is an additional parameter of the algorithm. Condition (3.29b) is taken from [20]. We still need a suitable set $\widehat{\mathcal{K}}_l$ in condition (2.10) of step 'Refine'. To this end, we show the following lemma.

Lemma 3.6 *For any $T\in\mathcal{T}_l$, there holds*

$$\nu_l(T)^q \leq\; C\left(\sum_{\substack{S\in\mathcal{S}_l\\ \text{with } S\subset\partial T}} \eta_l(S)^q + \sum_{\substack{\widetilde{T}\in\mathcal{T}_l\\ \text{with } \widetilde{T}=T \text{or } \widetilde{T}\cap T\in\mathcal{S}_l}} [\eta_l(\widetilde{T})^q + \delta_l(\widetilde{T})^q]\right). \tag{3.30}$$

Proof. We follow some lines of the proofs of [23, Propositions 1.14 and 1.15]. Let $T\in\mathcal{T}_l$; then

$$h_T\,\|f\|_{q;T} \;\leq\; h_T\,\|f_T\|_{q;T} \;+\; \delta_l(T).$$

Moreover, the fact $\|\!|\varphi_l^T|\!\|_T = 1$ implies $|T|^{1/q}\leq Ch_T^{-1}\int_T\varphi_l^T$ and so

$$\|f_T\|_{q;T}^q \;\leq\; Ch_T^{-1}|T|^{1/p}\int_T f_T\psi_T \tag{3.31}$$

for $\psi_T := |f_T|^{q-2}f_T\,\varphi_l^T$ (without loss of generality let $f_T\neq 0$). Noting

$$\int_\Omega a(\nabla u_l)\cdot\nabla\psi_T = 0$$

and

$$\|\psi_T\|_{p;T} \;\leq\; Ch_T|f_T|^{q-1}$$

by a Poincaré inequality, we obtain

$$\int_T f_T\psi_T = \int_\Omega f\psi_T + \int_T (f_T-f)\psi_T \leq C[\eta_l(T)+\delta_l(T)]|f_T|^{q-1}. \tag{3.32}$$

We combine (3.31), (3.32), and $|T|^{1/p}|f_T|^{q-1} = \|f_T\|_{q;T}^{q-1}$ as to arrive at

$$h_T\,\|f_T\|_{q;T} \;\leq\; C\,[\eta_l(T)+\delta_l(T)],$$

whence

$$h_T \, \|f\|_{q;T} \;\leq\; C \, [\eta_l(T) + \delta_l(T)]. \tag{3.33}$$

Similarly, making use of $|S|^{1/q} \leq C h_T^{-1/q} \int_S \varphi_l^S$,

$$\int_S [\![a(\nabla u_l) \cdot n]\!] \, \psi_S \;=\; -\int_\Omega a(\nabla u_l) \cdot \nabla \psi_S$$

for $\psi_S := |\, [\![a(\nabla u_l) \cdot n]\!] \,|^{q-2} \, [\![a(\nabla u_l) \cdot n]\!] \; \varphi_l^S$, we establish

$$\left\| h_S^{1/q} \, [\![a(\nabla u_l) \cdot n]\!] \right\|_{q;S} \leq C \left[\eta_l(S) + \sum_{T \in \mathcal{T}_l \text{ with } \partial T \supset S} h_T \, \|f\|_{q;T} \right] \tag{3.34}$$

for any $S \in \mathcal{S}_l$. Combing (3.33), (3.34), and consequences of (2.12) complete the proof.

*Remark 3.6 (*Equivalence) In addition to (3.30), the inequalities

$$\eta_l(T) \leq C \, h_T \, \|f\|_{q;T} \;\leq\; C \, \nu_l(T),$$

$$\eta_l(S) \leq C \left(h_S^{1/q} \, \|[\![a(\nabla u_l) \cdot n]\!]\|_{q;S}^q + \sum_{\substack{\widetilde{T} \in \mathcal{T}_l \\ \text{with } \partial \widetilde{T} \supset S}} h_{\widetilde{T}} \, \|f\|_{q;\widetilde{T}}^q \right)^{1/q}$$

$$\leq C \left(\sum_{\substack{\widetilde{T} \in \mathcal{T}_l \\ \text{with } \partial \widetilde{T} \supset S}} \nu_l(\widetilde{T})^q \right)^{1/q}$$

hold for any $T \in \mathcal{T}_l$ and any $S \in \mathcal{S}_l$. Therefore, the local error indicators (2.6) are equivalent (up to data oscillation) with the standard residual indicators $\nu_l(T), T \in \mathcal{T}_l$.

Invoking (3.30), we choose $\widehat{\mathcal{K}}_l$ as small as possible such that

$$\sum_{T \in \widehat{\mathcal{T}}_l} \nu_l(T)^q \;\leq\; C \sum_{K \in \widehat{\mathcal{K}}_l} \eta_l(K)^q. \tag{3.35}$$

Also this modification (which is basically the one of [19] for $p = 2$) of Algorithm 2.1 converges. In fact, by induction (see [20, Theorem 4.5]), one derives $\lim_{l \to \infty} \delta_l = 0$ from (3.29b) and the analogue of (3.11b). Furthermore, a slight modification of Proposition 3.1 gives

$$\lim_{l \to \infty} \left(\sum_{K \in \widehat{\mathcal{K}}_l} \eta_l(K)^q \right)^{1/q} = 0.$$

Refining and marking conditions (3.35) and (3.29a) therefore yield $\lim_{l\to\infty} \nu_l = 0$. As a consequence, consistency and then convergence follow from

$$\|\!|DJ(u_l)|\!\|_{\Omega;*} \;\leq\; C\,\nu_l,$$

which was proved by Baranger and El Amri [3, Sect. 2] (see also [23, Proposition 3.12]).

In a similar way, the convergence can be established for other local error indicators that are equivalent (up to data oscillation) to (2.6).

A comparison of the afore-mentioned error indicators would be out of the scope of this article. We restrict ourselves to the following few remarks. The computation of $\nu_l(T)$, $T \in \mathcal{T}_l$, requires slightly less mesh administration than the one of $\eta_l(K)$, $K \in \mathcal{K}_l$. However, the latter ones have the advantage that the marking for data oscillation needs not to be done independently. The computation of the indicators $\zeta_l(K)$, $K \in \mathcal{K}_l$, and $|\alpha_l(K)|$, $K \in \mathcal{K}_l$, is more expensive. But they may contain more 'nonlinear' information. In particular, $\zeta_l(K)$, $K \in \mathcal{K}_l$, seem to be attractive because of Remark 3.2, if one is interested in the approximation of the minimum value $J(u)$.

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References

1. M. Ainsworth and J. T. Oden, A posteriori error estimation in finite element analysis. John Wiley, New York, 2000
2. I. Babuška and M. Vogelius, Feedback and adaptive finite element solution of one-dimensional boundary value problems. Numer. Math. 44 (1984), 75–102
3. J. Baranger and H. El Amri, Estimateurs a posteriori d'erreur pour le calcul adaptatif d'écoulements quasi-newtoniens. RAIRO, Modélisation Math. Anal. Numér. 25 (1991), 31–48
4. J. W. Barrett and W. B. Liu, Finite element approximation of the p-Laplacian. Math. Comput. 61 (1993), 523–537
5. J. W. Barrett and W. B. Liu, Finite element approximation of the parabolic p-Laplacian. SIAM J. Numer. Anal. 31 (1994), 413–428
6. F. A. Bornemann, B. Erdmann, and R. Kornhuber, A posteriori error estimates for elliptic problems in two and three space dimensions. SIAM J. Numer. Anal. 33 (1996), 1188–1204
7. S. C. Brenner and L. R. Scott, The mathematical theory of finite element methods. Vol. 15 of Texts in Applied Mathematics, Springer, New York, 1994
8. S.-S. Chow, Finite element error estimates for nonlinear elliptic equations of monotone type. Numer. Math. 54 (1989), 373–393
9. P. G. Ciarlet, The finite element methods for elliptic problems. Vol. 4 of Studies in Mathematics and its Applications, North–Holland, Amsterdam, 1978

10. P. Clément, Approximation by finite element functions using local regularization. Revue Francaise Automat. Informat. Recherche. Operationelle Ser. Rouge Anal. Numér. 9 (1975), 77–84
11. P. Deuflhard, P. Leinen, and H. Yserentant, Concepts of an adaptive hierarchical finite element code. IMPACT Comput. Sci. Eng. 1 (1989), 3–35
12. W. Dörfler, A robust adaptive strategy for the non-linear Poisson equation. Computing 55 (1995), 289–304
13. W. Dörfler, A convergent adaptive algorithm for Poisson's equation. SIAM J. Numer. Anal. 33 (1996), 1106–1124
14. W. Dörfler and R. H. Nochetto, Small data oscillation implies the saturation assumption. Numer. Math. (2001), DOI 10.1007/s002110100321
15. W. Dörfler and O. Wilderotter, An adaptive finite element method for a linear elliptic equation with variable coefficients. ZAMM, 80 (2000), 481–491
16. R. Glowinski and A. Marrocco, Sur l'approximation, par éléments finis d'ordre un, et la résolution, par pénalisation-dualité, d'une classe de problèmes de Dirichlet non linéaires. R.A.I.R.O. 9 (1975), 41–76
17. P. Grisvard, Singularities in boundary value problems. No. 22 in Recherches en Mathematiques Appliquees, Paris: Masson. Berlin: Springer-Verlag, 1992
18. P. Jebelean, Finite-dimensional approximation and coerciveness in a problem with p-Laplacian. Nonlinear Anal., Theory Methods Appl. 33 (1998), 253–259
19. P. Morin, R. H. Nochetto, and K. G. Siebert, Data oscillation and convergence of adaptive FEM. SIAM J. Numer. Anal. 38 (2000), 466–488
20. P. Morin, R. H. Nochetto, and K. G. Siebert, Local problems on stars: a posteriori error estimators, convergence, and performance. Freiburg, Preprint No. 29, 2000, accepted for publication in Math. Comp.
21. M. Petzold, A posteriori error estimators for elliptic equations with discontinuous diffusion coefficients. Preprint No. 333, Weierstraß-Institut für Angewandte Analysis und Stochastik, 1999
22. A. Schmidt and K. G. Siebert, ALBERT: An adaptive hierarchical finite element toolbox. Preprint No. 06, Mathematische Fakultät Freiburg, 2000
23. R. Verfürth, A review of a posteriori error estimation and adaptive mesh-refinement techniques. Advances in Numerical Mathematics, John Wiley, Chichester, 1996
24. D. Wei, Finite element approximations of solutions to p-harmonic equation with Dirichlet data. Numer. Funct. Anal. Optimization 10 (1989), 1235–1251